

Code Quality Improvements

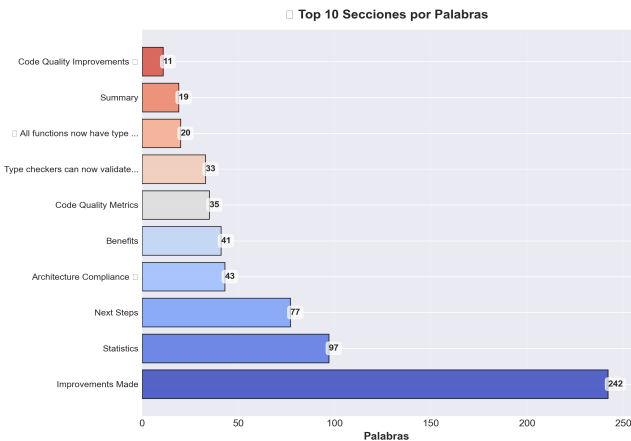
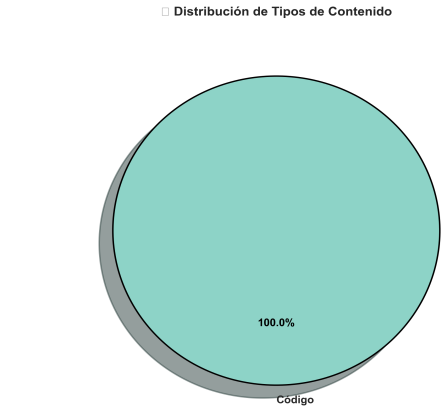
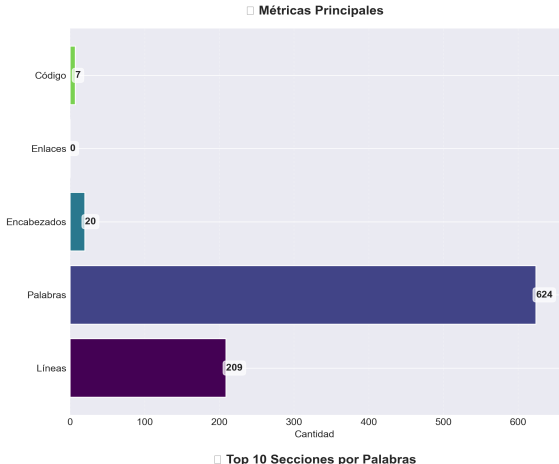
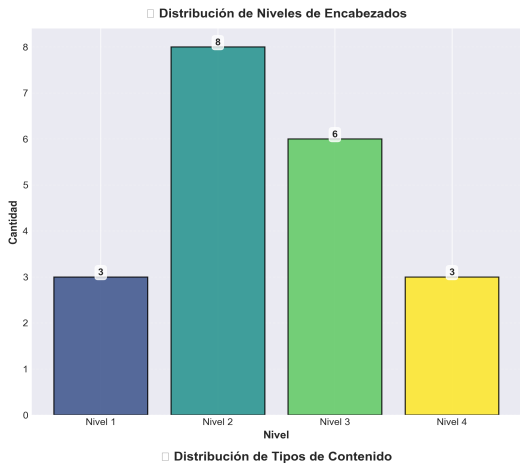
Análisis y Documentación Completa

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Análisis Visual Completo

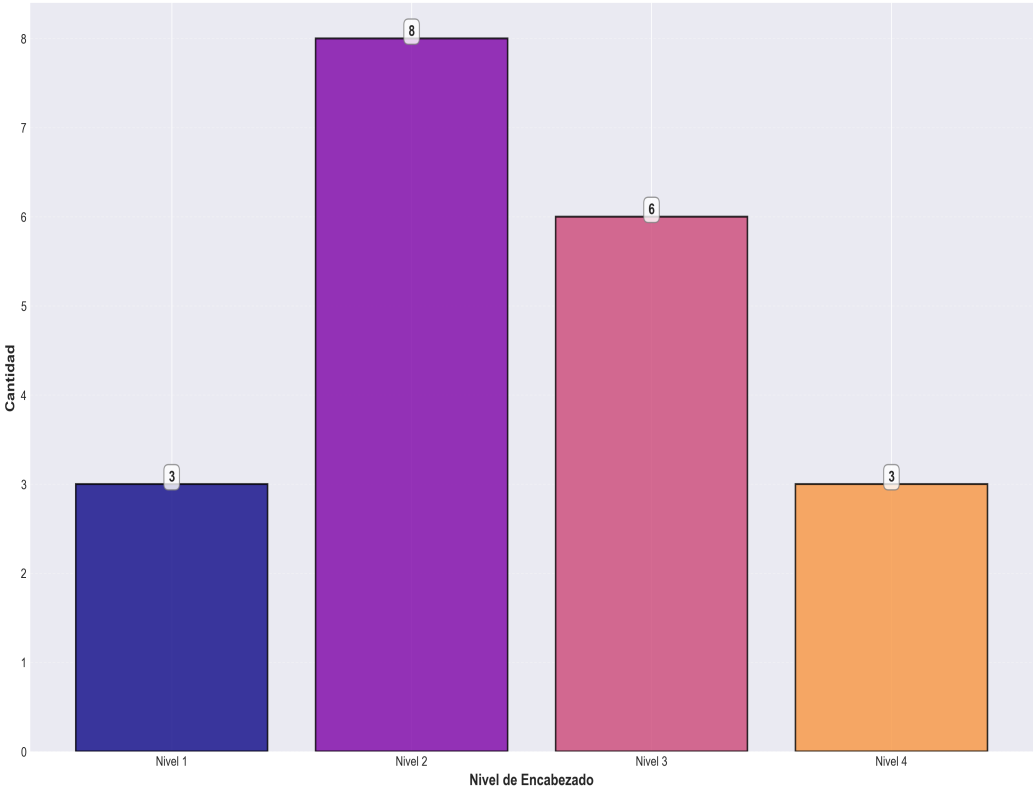
Dashboard Principal

Análisis Completo: CODE_QUALITY_IMPROVEMENTS

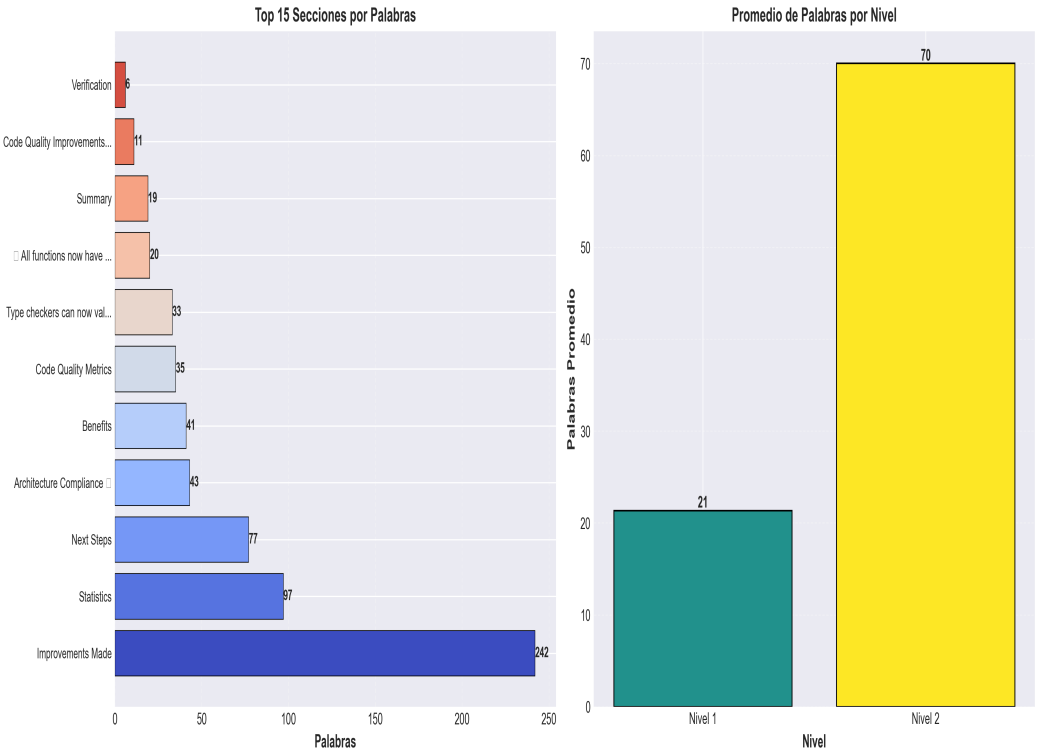


Análisis Adicionales

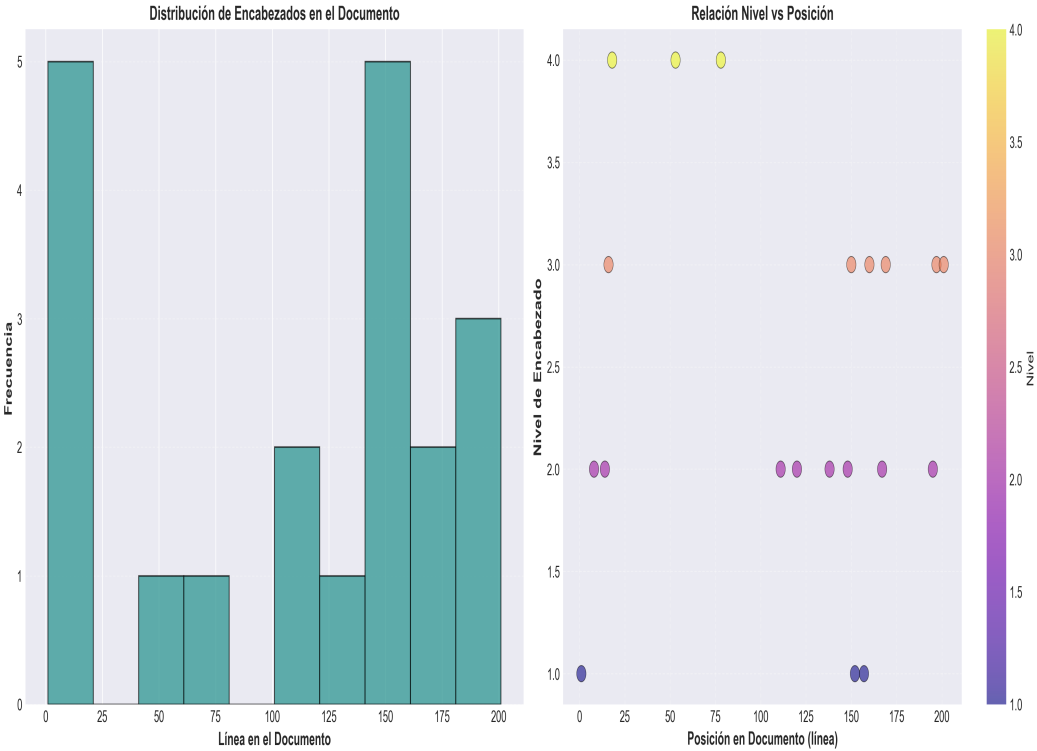
□ Estructura de Encabezados: CODE_QUALITY_IMPROVEMENTS



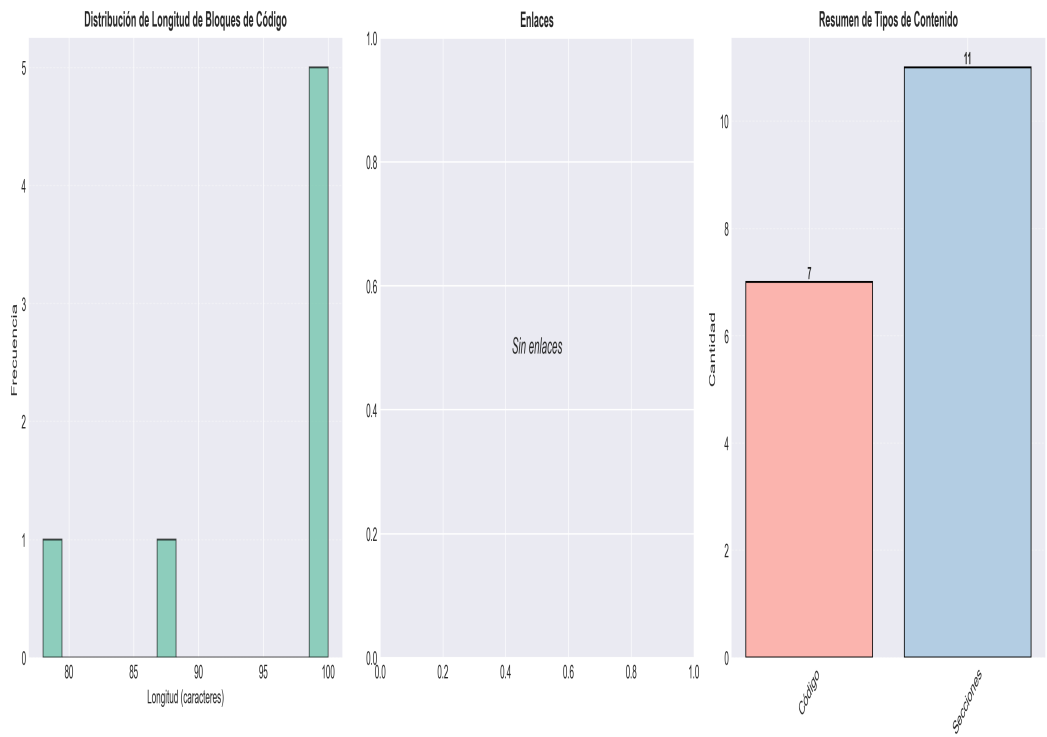
□ Análisis de Complejidad: CODE_QUALITY_IMPROVEMENTS



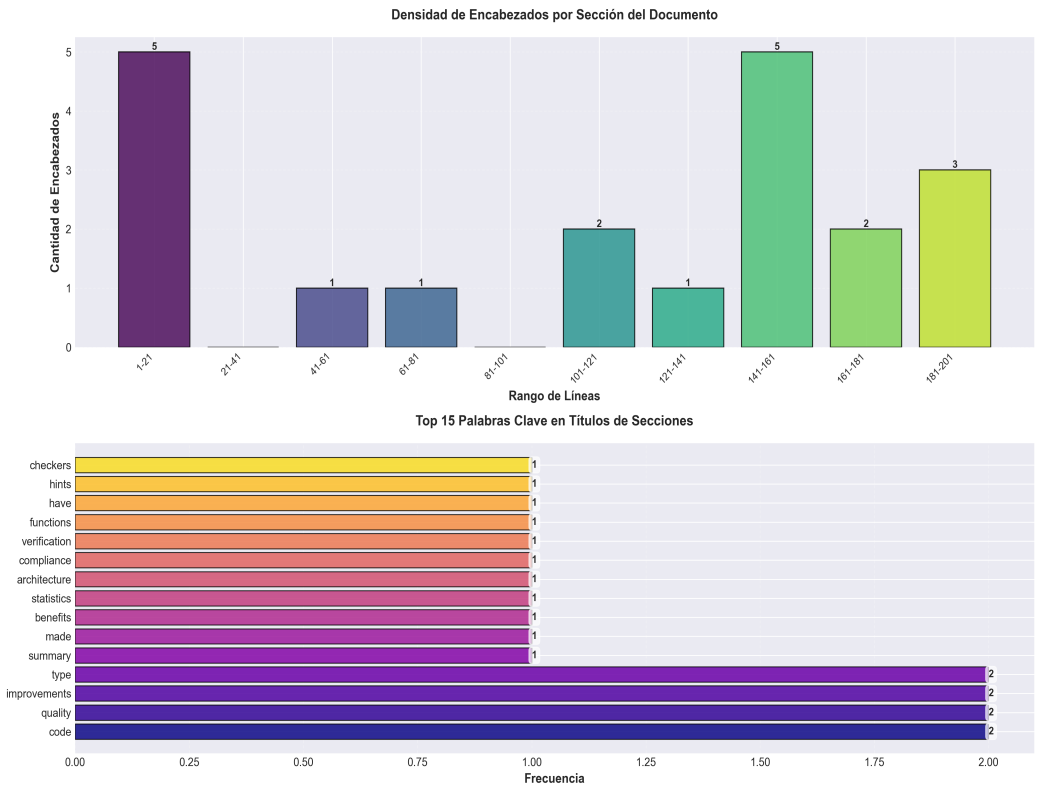
📊 Análisis Temporal y Distribución: CODE_QUALITY_IMPROVEMENTS



▣ Análisis Detallado de Contenido: CODE_QUALITY_IMPROVEMENTS



□
 Análisis de Densidad y Frecuencia: CODE_QUALITY_IMPROVEMENTS



Resumen Ejecutivo: Este documento contiene 624 palabras distribuidas en 209 líneas, con 20 encabezados organizados en 11 secciones principales.

Estadísticas del Documento

Métrica	Valor
Total de líneas	209
Total de palabras	624
Total de caracteres	5,244
Encabezados	20
Bloques de código	7
Tablas	0
Enlaces	0
Imágenes	0
Secciones principales	11

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Resumen del Contenido

Code Quality Improvements ■

****Date****: 2025-01-27

****Status****: ■ Completed

Summary

Implemented code quality improvements focusing on type hints, documentation, and code consistency across the API layer.

Improvements Made

1. Enhanced Type Hints ■

- ■ Added ``TYPE_CHECKING`` imports for forward references
- ■ Added type hints to ``initialize_services()`` function:
- ``pipeline``: "IntegratedPipeline"
- ``config_manager``: "ConfigManager"
- ``monitor``: "SystemMonitor"
- Return type: ``-> None``
- ■ Enhanced docstring with ``Raises`` section

****Before****

```
def initialize_services(pipeline, config_manager, monitor):  
    """Initialize all services."""
```

****After****

```
def initialize_services(  
    pipeline: "IntegratedPipeline",  
    config_manager: "ConfigManager",  
    monitor: "SystemMonitor"  
    ) -> None:  
    """
```

Initialize all services.

Args:

pipeline: IntegratedPipeline instance

config_manager: ConfigManager instance

monitor: SystemMonitor instance

Raises:

RuntimeError: If services are already initialized

"""

- ■ Added return type hint to `\_get\_document\_converter`:

- Return type: `-> "DocumentConverterAdvanced"`

- ■ Enhanced docstring with `Returns` and `Raises` sections

****Before:****

```
def _get_document_converter():
```

```
    """Get document converter instance."""
```

****After:****

```
def _get_document_converter() ->
```

```
    "DocumentConverterAdvanced":
```

```
    """
```

Get document converter instance.

Returns:

DocumentConverterAdvanced instance

Raises:

HTTPException: If document converter is not available

"""

- ■ Added return type hint to `store\_episode`:

- Return type: `-> MemoryStoreResponse`

- ■ Enhanced docstring with comprehensive `Args`, `Returns`, and `Raises` sections

****Before:****

```
async def store_episode(...):
```

```
    """Store episode in memory."""
```

****After:****

```
async def store_episode(...) -> MemoryStoreResponse:
```

.....

Store episode in memory.

Args:

request: Memory store request with episode data

req: FastAPI request object

service: Memory service instance

_auth: Optional API key for authentication

[Nota: El documento original tiene 209 líneas. Se muestra un resumen de las primeras 100 líneas.]